

Catalog of Forces

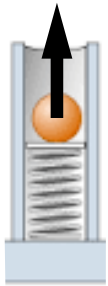
This is a list of the most common examples of forces you are likely to encounter in problems, along with the usual symbols used to identify them. If you are unable to describe a force in a problem as one of these, be cautious, you could be inventing up a force.



Gravitational Force (aka gravity, weight)

$$\vec{F}_G, \vec{F}_g, \vec{w}, m\vec{g}$$

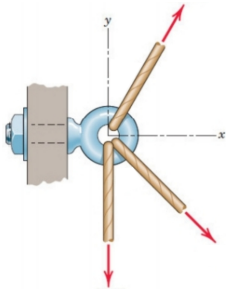
object with mass



Spring Force

$$\vec{F}_{sp}$$

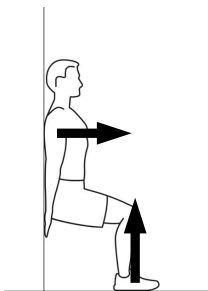
object in contact with spring



Tension

$$\vec{T}, \vec{F}_T, \vec{F}_t$$

object in contact with
rope, string, wire, line

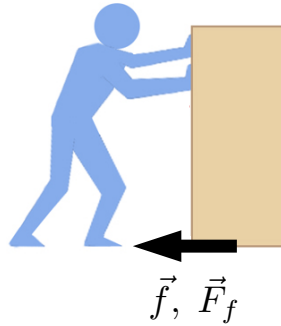


Normal Force

$$\vec{n}, \vec{F}_N, \vec{F}_n$$

surface pressing against
object; force perpendicular
to surface

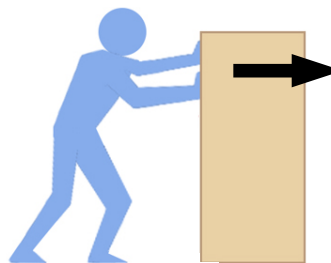
Friction and Drag (Air Resistance)



surfaces sliding or
tending to slide



object traveling
through air



Applied Forces

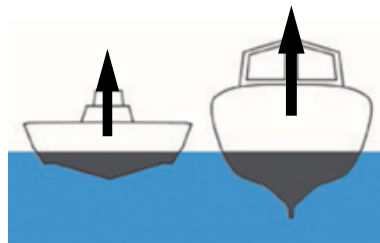
$$\vec{F}_p$$

push or pull on
object



Other Forces

Thrust



Buoyant Force